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# Clearance sale or not? Benefits from multi-behavioral factors of strategic customers☆

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ABSTRACT

We consider a seller's decision problem when facing a stochastic number of customers who are strategic. The seller first determines a strategy of either clearance sale or fixed price, and then stocks the corresponding quantity of products to maximize profit. Because customers potentially face a shortage if they wait for a discount, behavioral factors arise. Derived from prospect theory, risk aversion, reference point, and loss aversion are considered to capture the behavior of strategic customers. We investigate the behavioral impacts and find that customers' loss aversion always benefits the seller who chooses a clearance sale strategy, but risk aversion does not. The seller should also consider the cost of the product (or profit margin) to choose either a clearance sale or fixed-price. From the result of numerical analysis, the potential profit improvement from considering behavioral factors can be as high as 18.55% and is affected by the product cost and the discount rate. Furthermore, we point out that inaccurately estimating the behavioral factors may result in great profit losses, but when the product cost is high or the product cost is moderate with a high discount rate, a seller need not worry about the inaccuracy because it causes limited profit consequences.

## 1. Introduction

Should a retailer use clearance sales at the end of the selling season? As we know, both a clearance sale and a fixed price are common in practice. Customers are often lured by fashion brands such as Zara and H&M for their clearance sales; while warehouse wholesalers such as Costco and Sam Club are well-known for their fixed-price strategy. These two strategies cannot perfectly replace each other. The sale strategy found to be useful to one kind of sellers may not work for another or can even create great losses. For instance, J.C. Penney, a famous American branded retailer that once relied on clearance sales, changed its sale strategy to a fixed price throughout its stores in 2012 [1]. It then suffered steady declines in sales, ending up with a $4.3 billion loss in that year [2]. From the theoretical analysis of Su [3], products with low cost but high price should not be sold by the strategy of clearance sale. One cannot deny that the successes in sales are at least partially attributed to appropriate selections of sale strategy.

In the practice of clearance sale, customers have become so sophisticated that they do not always buy immediately but also strategically delay their purchasing time until there is a discount [4,5]. Hence, when thinking about clearance sale, the influence of strategic customer behavior should be taken into account. Because of the strategic customer behavior, clearance sales ultimately result in a decline in the demand during the main selling period and remarkable retail sales transacted at discounted prices. Thus, most of the prior studies have argued that charging a fixed price would do a better job than that of a clearance sale [6,7]. However, this argument is based on assumption that customers are fully rational in determining their time of purchase, but actually they are not.

Several empirical and experimental studies provide evidence that strategic customers have decision biases and behavior anomalies [8,9]. Some of them consider regret or disappointment which actually come from perceived losses [10,11], but rare studies take into account loss aversion. We believe that loss aversion based on the reference point of the status quo is more likely to be the underlying reason of regret or disappointment which should be mainly considered. Moreover, most of the theoretical studies only consider one behavioral factor, but how about the integrated effects of multiple behavioral factors? Accordingly, we build an integrated behavioral model for strategic customers based on Prospect Theory with full consideration of reference points, risk aversion, and loss aversion to explore the mutual restriction among various behavior.

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We study the impact of strategic customers on selection of the sale strategy and especially the salient role of customer behavioral factors. The behavioral factors of customers are adopted to address the following questions: (i) How does the presence of behavioral factors affect customers' choices and the seller's decisions? (ii) Does the seller benefit or suffer from these behavioral factors? (iii) How much will the seller potentially gain if he incorporates risk aversion and loss aversion? How about the potential gain when misperceiving them? (iv) What role do the product cost and the discount rate play in affecting the seller's decisions? We find that although loss aversion always incurs more buying-immediately decisions, and thus favor the adoption of a clearance sale strategy, risk aversion does not, depending on the product cost and the discount. As a result, the seller would benefit from loss aversion but not always from risk aversion. Incorporating the behavioral factors can generate substantial profit improvements. The potential profit improvement is affected by the product cost and the discount rate and can be as high as % on average. In addition, whether inaccurately estimating the behavioral factors causes great profit losses are also associated with the product cost and the discount rate. For example, in case of a high product cost or a moderate cost but a high discount rate, the inaccuracy causes limited profit consequences.

There are three contributions to the literature. First, we are the first to investigate the joint impacts of reference point, risk aversion, and loss aversion of strategic customers on the decisions of the seller. Since the demand is stochastic and customers are heterogeneous, all these behavioral factors are playing roles in customers' purchasing decisions. Second, our study brings new insights in when to adopt different sale strategies. We demonstrate that loss aversion always favors a clearance sale, but whether or not risk aversion favors a clearance sale depends on the product cost and discount rate. Finally, we quantify the potential profit improvements when incorporating and accurately estimating the behavioral factors. In particular, the product cost and discount rate may weaken the potential behavioral benefits if any. Hence, the seller needs to discreetly evaluate the behavioral benefits depending on the cost and discount parameters of its products, which in turn guides the choice of the specific sale strategy.

The rest of this paper is organized as follows. In Section 2, we review the related literature. In Section 3, we model strategic customers' behaviors. In Section 4, we focus on the decisions of the seller and describe the optimality of two sale strategies. In Section 5, we analyze the stock quantity and profit impacts of behavioral factors and how the behavioral impacts interact with product cost and the discount rate. In Section 6, we consider the sensitivity of the seller's profits to price and stock quantity decisions. Finally, we summarize the conclusions in Section 7. All proofs are shown in the appendix.

## 2. Literature review

The prevalence of strategic customers has been attracting increased attention from academia in a wide range of fields, from economics [12] to marketing [13] to operations management [14–16]. In recent years, several researchers have made contributions to this issue [17–19]. Papanastasiou and Savva [20] consider the dynamic pricing problem when the retailer faces strategic customers with social learning. Hao et al. [21] develop a framework where a high-tech firm sequentially introduces new products with incremental innovation over two selling seasons in the presence of strategic consumers and uncertain demand. Farshbaf-Geranmayeh and Zaccour [22] develop a multi-period model to investigate pricing and advertising decisions in a supply chain with considering myopic and strategic customers. Chen et al. [23] consider a seller who can use the strategic customer's purchase history to learn the customer's valuation distribution and make pricing decisions to maximize her long-run revenue. Yang et al. [24] study the influence of strategic customers on the co-opetition strategy between a manufacturer of a proprietary component and an original equipment manufacturer.

A growing body of research suggests that strategic customers are not perfectly rational [25]. Several empirical and experimental studies provide evidence that strategic customers have decision biases and behavior anomalies. Song et al. [9] showed that customers are affected by bounded rationality and risk aversion when making their buying or waiting decisions. Baucells et al. [8] suggested that strategic customers are affected by risk aversion, hyperbolic discounting, and sub-endurance. The behavioral factors affect not only the optimal stock quantity but also the sale strategy. For example, Liu and van Ryzin [26] found that the clearance sale strategy is not profitable when customers are risk-neutral, but high levels of risk aversion tend to favor this strategy as an optimal one. Özer and Zheng [10] found that customer behaviors accounted for availability misperception and regret, and reported that the behavioral factors reinstated the profitability of the markdown strategy over the fixed-price one. The debate about the impact of behavior of strategic customers continues [27–29].

In this study, we revisit the question of which the two sale strategies (fixed-price and clearance sale) is better. Different from the previous study, we consider a seller facing a stochastic number of strategic customers from a behavioral perspective. Specifically, the seller selects either of the two sale strategies and optimizes the stock quantity of products, and then, customers choose between buying and waiting with decision biases. That may pose a question: which behavioral factors should be considered for the decision biases of customers? [26] consider risk aversion and suggest that the seller will be benefited more by creating a rationing risk in two period selling problems. Özer and Zheng [10] emphasize regret when a customer initially chooses to wait but encounters stockout later, or when the consumer buys the product at a high price but realizes that the product is still available at the markdown price. Ying Zhang [30] and Xu and Duan [11] think that strategic customers are disappointment averse if the outcome is worse than their prior expectation.

The existing studies support that the behavioral factors are the reasons of decision biases which are from the decision-making process. Accordingly, risk preferences and perceived probabilities exist in the uncertain decision-making problems [31,32], but [9] have conducted an experiment and provided evidence that the influence of perceived probabilities on strategic customers is not significant. Hence, instead of perceived probabilities, risk preferences should be considered. On the other hand, the full price purchase (period 1) is the status quo for strategic customers because they make inter-temporal decisions. Hence, bias toward the status quo should be considered [33,34]. That is, the full price purchase is a reference point, and when customers failed in the discount period, the condition of full price purchase will naturally cause the perceived loss. Therefore, the loss aversion based on the reference point should be considered [33,35]. Several studies consider regret or disappointment which actually come from perceived losses [31,32], but rare studies take into account loss aversion. We believe that loss aversion based on the reference point of the status quo is more likely to be the underlying reason of regret or disappointment which should be mainly considered. Moreover, most of the theoretical studies only consider one behavioral factor, but rare of them consider the integrated effects of multiple behavioral factors. Hence, we should attach more importance to an integrated behavioral model for strategic customers based on Prospect Theory with full consideration of reference points, risk aversion, and loss aversion.

## 3. The strategic customer behavior

In this section, we will build the decision model of strategic customers and discuss the behavioral factors of customers when they make strategic decisions.

### 3.1. Model setup

We consider a retailer in which a monopolistic seller sells a single product to a group of customers that may behave strategically. The

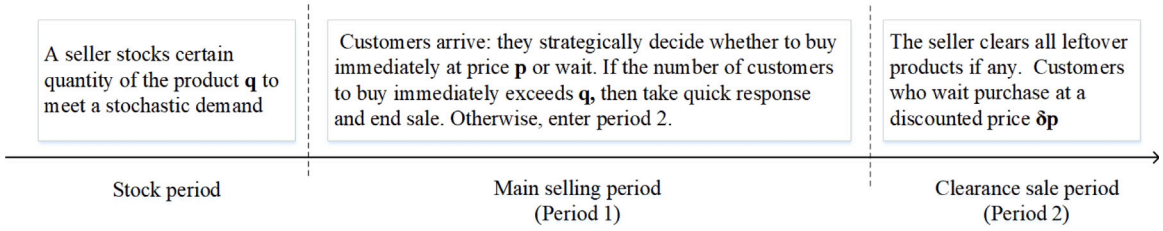



**Fig. 1.** Sequence of events under the clearance sale strategy.

number of customers $X$ is a random variable that follows a probability distribution with cdf. $F(\cdot)$ and pdf. $f(\cdot)$. The retail price denoted by $p$ is exogenous. The selling season lasts for two periods. The seller first chooses between two sale strategies: fixed-price for two periods or clearance sale in the last period. Under fixed-price strategy, the seller determines the optimal stock quantity $q$ at a unit cost of $c$ and sells to the customers at the same price $p$ through the selling season. Under the clearance sale strategy, the seller optimizes the stock quantity $q$ before the selling season and sells to the customers at price $p$ in period 1, then clears the leftover products (if any) at a discounted price $\delta p$ in period 2, where $\delta$ is an exogenous discount factor [36].

Given the seller's decisions, customers decide whether and when to buy the product. Each customer buys at most one unit of product. The value $v$ of a product is private information for each customer, but it is common knowledge that $v$ is uniformly distributed on $[0, \bar{v}]$. Such an assumption is usually used in the literature (e.g., 36,37). Let $G(\cdot)$ denote the cumulative distribution function of $v$, and then $\bar{G}(\cdot) = 1 - G(\cdot)$.

Every customer who chooses to buy at full price $p$ is guaranteed to have the product, even when the buying customers are more than $q$. The excess demand can be satisfied by replenishment as a quick response strategy [36,38,39]. Replenishment is usually highly costly, even resulting in little or no profit. Hence, the seller is only motivated to replenish products with high selling prices to ensure the service level, but not motivated to do replenishment in period 2 for creating a possible stockout to induce customers to buy earlier. For the sake of simplicity, we assume that the replenishment cost is equal to the selling price $p$, i.e., the profit from excess demand is zero. Moreover, all leftovers after period 2 are disposed of at zero.

Under the fixed-price strategy, a customer buys the product as long as $v > p$. As such, the seller's stock quantity decision is typically a newsvendor problem. However, under the clearance sale strategy, a customer should strategically decide either to purchase at full price $p$ with sure availability or wait for the discount price $\delta p$ but with possible stockout. The decision process of the clearance sale is depicted in Fig. 1. For the decision process of strategic customers in clearance sale is complex, we first model the strategic customer behavior in clearance sale and formalize the behavioral factors of strategic customers. And then, the seller's optimal stock quantity decision and the ensuing profit are analyzed. Furthermore, we compare the decisions and profits between the fixed-price and clearance sale strategies to derive the seller's optimal sale strategy. Finally, we quantify the behavioral benefit in terms of profit improvements in clearance sale and review the choice between the two sell strategies.

### 3.2. The decision of strategic customers

We first build a baseline model under the clearance sale to capture strategic customer decisions ignoring behavioral factors. Recall that a customer has a heterogeneous value $v$ for the product over two periods. If he/she chooses to buy the product in period 1, the customer obtains a utility $U_b(v) = v - p$. If he/she chooses to wait for a discount until period 2, the customer obtains a utility $U_w(v) = (v - \delta p)\phi$, where $\phi \in [0, 1]$ denotes the product availability for a waiting customer (i.e., fill rate) and is determined by the seller's stock quantity of the product $q$ and other customers' decisions. To compute the fill rate, we define a cutoff value $v^*$ at which a customer is indifferent in buying and waiting, that is, $U_b(v^*) = U_w(v^*)$. It is easy to see that $v^* > p$. For a customer with $v < v^*$, waiting is the optimal choice; for a customer with $v > v^*$, buying is the optimal choice. Then, $\bar{G}(v^*)$ corresponds to the proportion of customers who choose to buy, whereas, $\bar{G}(\delta p) - \bar{G}(v^*)$ corresponds to the proportion of customers who wait to buy the leftovers. Thus, the number of leftover products is $[q - X\bar{G}(v^*)]^+$, where $x^+ = \max(x, 0)$. As such, the fill rate is given by the ratio of the leftover quantity to the number of waiting customers and is written as follows,

$$\phi = E\left[\frac{[q - X\bar{G}(v^*)]^+}{X[\bar{G}(\delta p) - \bar{G}(v^*)]} \wedge 1\right] = F\left(\frac{q}{\bar{G}(\delta p)}\right) + \int_{\frac{q}{\bar{G}(\delta p)}}^{\frac{q}{\bar{G}(v^*)}} \frac{q - x\bar{G}(v^*)}{x[\bar{G}(\delta p) - \bar{G}(v^*)]} f(x)dx. \tag{1}$$

where $x \wedge y = \min(x, y)$. For simplification of calculation, $F(\cdot)$ and $f(\cdot)$ are assumed to be continuous. By the definition of the cutoff value $v^*$, we have $U_b(v^*) = U_w(v^*)$, i.e., $v^* - p = (v^* - \delta p)\phi$. Hence, we can also express the fill rate as,

$$\frac{v^* - p}{v^* - \delta p} = \phi. \tag{2}$$

For a given stock quantity $q$, $v^*$ and $\phi$ can be obtained by solving Eqs. (1) and (2). However, there may be no solutions for the equations in some cases. The existence and monotonicity of $v^*$ is given in Proposition 1.

**Proposition 1.** *(a) The cutoff value $v^*$ exists if and only if the following condition is satisfied: $\frac{q}{\bar{G}(\delta p)} \int_{\frac{q}{\bar{G}(\delta p)}}^{+\infty} \frac{F(x)}{x^2} dx \le \frac{\bar{v} - p}{\bar{v} - \delta p}$. If this condition is not met, then the cutoff value $v^*$ does not exist, and the best choice for customers is to wait for a discount.*

*(b) Both cutoff value $v^*$ and the fill rate $\phi$ increase as the parameter $q$ increases.*

Appendix A presents the proofs of all the propositions. Proposition 1 describes the condition that $v^*$ exists and the monotonicity of $v^*$ and $\phi$ with respect to $q$. Specifically, Part (a) of the proposition implies that for any $q$ satisfying $\frac{q}{\bar{G}(\delta p)} \int_{\frac{q}{\bar{G}(\delta p)}}^{+\infty} \frac{F(x)}{x^2} dx \le \frac{\bar{v} - p}{\bar{v} - \delta p}$, the fill rate satisfies that $\phi(q, v^*) < \frac{\bar{v} - p}{\bar{v} - \delta p}$. This means that both cases of buying and waiting coexist in the system, i.e., the cutoff value $v^*$ exists. However, in the case that $\frac{q}{\bar{G}(\delta p)} \int_{\frac{q}{\bar{G}(\delta p)}}^{+\infty} \frac{F(x)}{x^2} dx > \frac{\bar{v} - p}{\bar{v} - \delta p}$, the fill rate satisfies that $\phi(q, v^*) > \frac{\bar{v} - p}{\bar{v} - \delta p}$, which implies that there is no cutoff value. In such cases, the cutoff value can be seen as $\bar{v}$, which means that waiting for a discount is always the best choice. Part (b) of the proposition reflects the impact of the stock quantity on the equilibrium behavior of strategic customers. When stock quantity $q$ becomes large, the number of buying customers becomes small (i.e., the cutoff value $v^*$ gets large), and the fill rate becomes high.

Fig. 2 presents a numerical example for Proposition 1 with $\bar{v} = 120$, $p = 50$, $\delta = 0.5$, $v \sim$ unif$[0, \bar{v}]$, and $x \sim$ unif$[0, 100]$. The figure confirms that $v^*$ exists only when $q$ is relatively small; the fill rate and the cutoff value increase with $q$. It is worth mentioning that the existence of $v^*$ implies fewer customers buying at the retail price $p$ than that under the fixed-price strategy. This fact implies the negative impact of strategic customers for the seller. It is plausible for the seller to deliberately

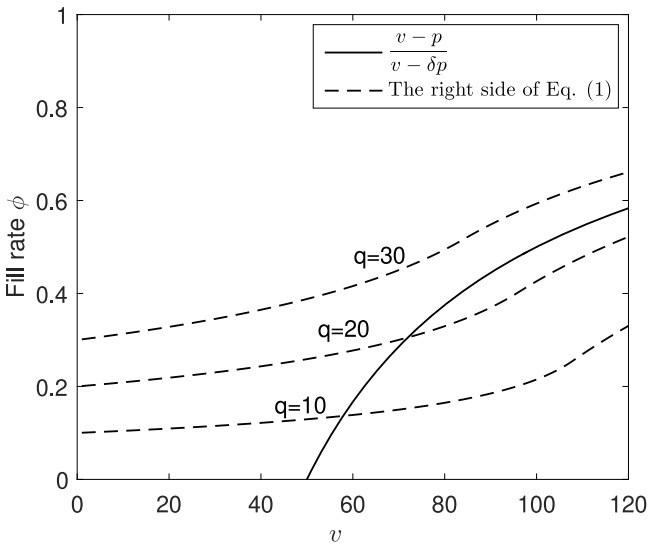



**Fig. 2.** Equilibrium solutions ($v^*$, $\phi$) with varying $q$.

decrease her stock quantity on discourage strategic customers from waiting for a discount [40]. Further more, the limited stock quantity brings more rationing risk, and thus, the behavior of strategic customers cannot be ignored under the uncertain decision circumstance [26]. In what follows, we attempt to calibrate the above model of strategic customers by incorporating the behavioral factors of Prospect Theory.

### 3.3. The behavior of strategic customers

A customer evaluates the economic outcomes in reliance on the reference point [31,41]. More often, they anchor on the current status [34], that is, customers tend to take the status quo as a reference point, and changes deviating from the status quo will be regarded as gains and losses. In our model, the guaranteed payoff from buying at a full price would then be taken as a reference point [42]. As such, given a certain payoff $v-p$, if a customer chooses to wait and obtains a product successfully, his perceived payoff is a gain of $(v-\delta p)-(v-p)$ $(=p-\delta p)$; conversely, if he obtains nothing, his perceived payoff is a loss of $-(v-p)$.

Denote $r$ as the perceived payoff of a customer, and thus we have $r=p-\delta p$ if the customer obtains a product by waiting, $r=-(v-p)$ if the customer obtains nothing. Based on Function (5) of Prospect Theory in [32], we build the behavioral utility function of strategic customers who choose to wait as a two-part power function of the form:

$$u(r)=\begin{cases} r^{\gamma}, & if \quad r\geq 0 \\ -\lambda\cdot(-r)^{\gamma}, & if \quad r<0 \end{cases} \tag{3}$$

It is a bit different from [32]. We give the same risk-averse parameter $\gamma$ $(0<\gamma<1)$ to gains and losses, because even though Tversky and Kahneman [32] present different risk parameters to gains and losses, they have found that both of the two medians of parameters are 0.88 by experiment. Moreover, a power utility function is a common form to describe risk aversion in the economics literature and is also widely used in the studies of strategic customer behavior [9,26]. As to modeling loss aversion that people are more sensitive to losses than to gains, we also follow [32] to set $\lambda(\geq 1)$ as a loss-averse parameter, representing the extent of loss aversion. Such a piecewise linear loss-averse utility function is widely used in behavioral literature [37,43].

Together with the two part power function associated with behavioral factors, we formalize the utilities of buying and waiting for strategic customers. If a customer buys the product immediately in period 1, the perceived expected utility is $\widehat{U}_b(v)=0$. If a customer chooses to wait, the perceived expected utility of waiting is $\widehat{U}_w(v)=(p-\delta p)^{\gamma}\hat{\phi}-\lambda(v-p)^{\gamma}(1-\hat{\phi})$, where $\hat{\phi}$ denotes the behavioral analogue of the fill rate $\phi$, and can definitely be computed by Eq. (1). A customer will not hesitate to buy the product when $\widehat{U}_w(v)<\widehat{U}_b(v)=0$, but wait otherwise. Recall the indifference condition between buying and waiting, we have,

$$\frac{\lambda(\hat{v}^*-p)^{\gamma}}{(p-\delta p)^{\gamma}+\lambda(\hat{v}^*-p)^{\gamma}}=\hat{\phi}, \tag{4}$$

where $\hat{v}^*$ is a behavioral analogue of the cutoff value. As seen, setting $\gamma=\lambda=1$, the above equation then degenerates into the case that no behavioral factors are at play, see Eq. (2). Solving Eqs. (1) and (4) yields the equilibrium solutions $(\hat{v}^*,\hat{\phi})$, Parallel with part (a) of Proposition 1, we give the condition that ensures the behavioral cutoff value $\hat{v}^*$ to exist.

**Proposition 2.** *(a) The cutoff value $\hat{v}^*$ exists if and only if the following condition is satisfied: $\frac{q}{\bar{G}(\delta p)}\int_{\frac{q}{\bar{G}(\delta p)}}^{+\infty}\frac{F(x)}{x^2}dx\leq\frac{\lambda(\bar{v}-p)^{\gamma}}{(p-\delta p)^{\gamma}+\lambda(\bar{v}-p)^{\gamma}}$. If this condition is not met, then the cutoff value $\hat{v}^*$ does not exist, and the best choice for customers is to wait for a discount.*

*(b) Both cutoff value $\hat{v}^*$ and the fill rate $\phi$ increase as the parameter $q$ increases.*

Proposition 2 indicates that even introducing the risk aversion and loss aversion, the properties of the fill rate and the cutoff value remain unchanged regarding the stock quantity, that is, part (b) of Proposition 1 still holds. The following proposition presents these changes with the parameters of risk aversion $\gamma$ and loss aversion $\lambda$.

**Proposition 3.** *(a) The cutoff value $\hat{v}^*$ is differently depending on the value of $q$ in relation to a threshold $\bar{q}(p,\bar{v},\delta,\lambda)$. Specifically:*

- *If $q<\bar{q}(p,\bar{v},\delta,\lambda)$, then $\hat{v}^*$ increases as $\gamma$ increases.*
- *If $q>\bar{q}(p,\bar{v},\delta,\lambda)$, then $\hat{v}^*$ decreases as $\gamma$ increases.*

*The threshold $\bar{q}(p,\bar{v},\delta,\lambda)$ is defined by:*

$$F\left(\frac{q}{\bar{G}(\delta p)}\right)+\int_{\frac{q}{\bar{G}(\delta p)}}^{\frac{q}{\bar{G}(2p-\delta p)}}\frac{q-x\bar{G}(2p-\delta p)}{x[\bar{G}(\delta p)-\bar{G}(2p-\delta p)]}f(x)dx=\frac{\lambda}{1+\lambda};$$

*(b) The cutoff value $\hat{v}^*$ decreases as $\lambda$ increases.*

Proposition 3 shows that the behavioral cutoff value $\hat{v}^*$ becomes smaller than that of the standard model if customers are loss-averse, that is, $\hat{v}^*<v^*$, and thus, the (perceived) fill rate is lower in accordance with Proposition 1. However, if strategic customers are risk averse, the behavioral cutoff value $\hat{v}^*$ becomes smaller than that of the standard model conditionally, i.e., if $q>\bar{q}(p,\bar{v},\delta,\lambda)$, the (perceived) fill rate is lower. It implies that creating rationing (fill rate is smaller than 1) may benefit the seller because behavioral customers tend to choose to buy at full price to avoid the possible risk of stockout in period 2. This result is just the same as the literature, Liu and van Ryzin [26] and Zhang and Cooper [6] also suggest that whether rationing is necessary and what level of rationing risk should be created depend on the capacity, pricing policy, and other characteristics of products. We are interested in how much profit improvement the seller can earn attributed to the two behavioral biases of strategic customers. Moreover, since the discount rate $\delta$ impacts the fill rate and the utility of waiting, will an increased $\delta$ reinforce or mitigate such behavioral benefits?

## 4. The decisions of the seller

In this section, a fixed-price strategy and a clearance sale strategy are presented and the comparison of the two selling strategies is discussed.

### 4.1. A fixed-price strategy

We first consider a fixed-price strategy. Under this strategy, the seller sells her product at an identical price $p$. A customer buys the

product if his/her value for the product is larger than $p$. Hence, the total expected demand for any given price $p$ is $E[\bar{G}(p)X]$. The seller should optimally set her stock quantity to meet a random demand. As such, the seller's decision problem can be formulated to maximize her expected profit by choosing $q_F$, which is,

$$\max\ \Pi_F(q_F) = pE\left[\bar{G}(p)X \wedge q_F\right] - cq_F. \tag{5}$$

In essence, the above problem that the seller optimizes the stock quantity under the fixed-price is a newsvendor problem. Let $q_F^*$ be the optimal stock quantity and $\Pi_F^*$ be the maximum expected profit. The optimal solution $q_F^*$ is then calculated by $F(\frac{q_F^*}{\bar{G}(p)}) = 1 - \frac{c}{p}$. With $q_F^*$, we can easily get $\Pi_F^*$ from Eq. (5).

### 4.2. A clearance sale strategy

Under a clearance sale, the seller sells the product in period 1 at price $p$ and the leftover products that are carried over to period 2 are cleared at a discount price $\delta p$. Customers have to estimate the fill rate at the beginning of the selling season and choose whether to buy or to wait based on their own $v$. The seller makes the stock quantity decision $q$ with a belief of $\hat{v}^*$, which is solved by Eq. (4). Products are guaranteed to be available to the customers who buy at full price in the selling season, thus excess demand is met for customer satisfaction but ends up with no profit. In the seller's mind, the stock quantity $q$ can induce the customers with $v > \hat{v}^*$ to buy immediately, and leftover products (if any) meet the waiting customers. Then, the seller's problem is to decide $q$ to maximize her profit as follows,

$$\begin{aligned} &\max\ \Pi(q,\hat{v}^*) = pE\left[\bar{G}(\hat{v}^*)X \wedge q\right] + \delta pE\left[(q-\bar{G}(\hat{v}^*)X)^+ \wedge \left(G(\hat{v}^*) - G(\delta p)\right)X\right] - cq, \\ &s.t. \qquad \frac{\lambda(\hat{v}^*-p)^\gamma}{(p-\delta p)^\gamma + \lambda(\hat{v}^*-p)^\gamma} = \hat{\phi}. \end{aligned} \tag{6}$$

The constraint specifies the cutoff value and the fill rate in the presence of behavioral factors. Let $(q^*, \hat{v}_0^*)$ be the joint solution and $\Pi^*$ be the maximum profit of optimization problem. The optimal solution $q^*$ is then calculated by maximizing the objective function $\Pi(q)$ while considering $\hat{v}^*$ in the constraint. That is, $(q^*, \hat{v}_0^*)$ can be calculated by the first derivative of $\Pi(q,\hat{v}^*)$ and the constrained equation of the optimization problem (6).

**Proposition 4.** *The joint solution $(q^*, \hat{v}_0^*)$ satisfies the following conditions:*

$$(a)\ \frac{p-c}{p} = (1-\delta)F(\frac{q}{\bar{G}(\hat{v}^*)}) - \delta F(\frac{q}{\bar{G}(\delta p)}) + (1-\delta)\frac{\partial \bar{G}(\hat{v}^*)}{\partial q}\int_0^{\frac{q}{\bar{G}(\hat{v}^*)}} x\mathrm{d}F(x),$$

$$(b)\ \frac{\lambda(\hat{v}^*-p)^\gamma}{(p-\delta p)^\gamma + \lambda(\hat{v}^*-p)^\gamma} = F(\frac{q}{\bar{G}(\delta p)}) + \int_{\frac{q}{\bar{G}(\delta p)}}^{\frac{q}{\bar{G}(\hat{v}^*)}} \frac{q - x\bar{G}(\hat{v}^*)}{x[\bar{G}(\delta p) - \bar{G}(\hat{v}^*)]} f(x)dx.$$

Proposition 4(a) indicates that for a give $\hat{v}^*$ the optimal solution $q^*$ can be derived by solving the equation. Proposition 4(b) suggests that for a given $q$, the corresponding $\hat{v}_0^*$ can be derived by solving the equation. The behavioral parameters risk aversion $\gamma$ and loss aversion $\lambda$ only present on the impact of risk aversion $\gamma$ and loss aversion $\lambda$, but they influence $q^*$ because they influence $\hat{v}^*$. Furthermore, given the impact of risk aversion $\gamma$ and loss aversion $\lambda$, the seller's optimal stock quantity and the ensuing profit may increase or decrease due to the decision bias of strategic customers. We give numerical examples for the joint solution in Fig. 3. Fig. 3 shows the curves of Proposition 4(a) in black and 4(b) in green with the same parameters in Fig. 2. The intersections of Proposition 4(a)'s curves and Proposition 4(b)'s curves are the joint solutions $(q^*, \hat{v}_0^*)$ with different cost ($c = 25, 30, 35$) and behavioral parameters ($\lambda = 1, 1.4, 1$ and $\gamma = 1, 1, 0.7$).

The Proposition 4(a)'s curves in Fig. 3 show that the intersections are decreasing with the increase of cost $c$, i.e., the joint solutions $(q^*, \hat{v}_0^*)$ decrease in $c$. The Proposition 4(b)'s curves in Fig. 3 show that loss aversion results in higher $q^*$ and lower $\hat{v}_0^*$, but the effect of risk aversion are more complex. The curve with risk aversion $\gamma = 0.7$ intersect

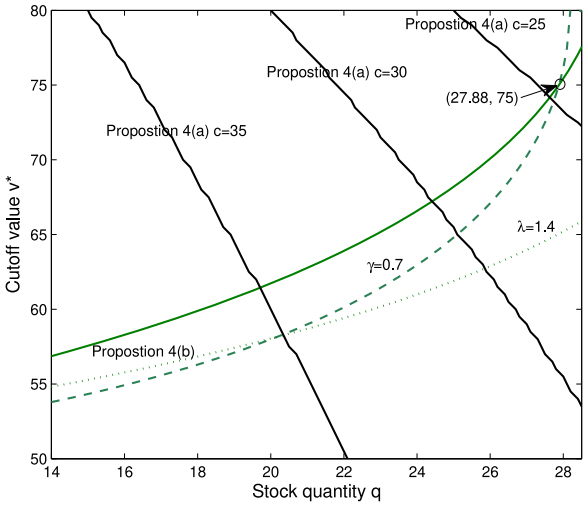



**Fig. 3.** The optimal solution changes with different parameters (with $\delta = 0.5$). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

with the curve with $\gamma = 1$ at the point $(q = 27.88, v^* = 75)$ which equals to $(\bar{q}(p,\bar{v},\delta,\lambda), 2p - \delta p)$. This intersection is the turning point of the influence of risk aversion which is consistent with the prediction of Proposition 3. Hence, the monotonicity of the joint solutions and profit with respect to the two behavioral parameters can be given as Proposition 5 in accordance with Propositions 3 and 4.

**Proposition 5.** *(a) The joint solutions $(q^*, \hat{v}_0^*)$ and profit $\Pi^*$ exhibit different patterns depending on the value of $c$ relative to a threshold $\bar{c}(p,\bar{v},\delta,\lambda)$. Specifically:*

- *When $c > \bar{c}(p,\bar{v},\delta,\lambda)$, both $(q^*, \hat{v}_0^*)$ and $\Pi^*$ increase as $\gamma$ increases.*
- *When $c < \bar{c}(p,\bar{v},\delta,\lambda)$, both $(q^*, \hat{v}_0^*)$ and $\Pi^*$ decrease as $\gamma$ increases.*

*The threshold $\bar{c}(p,\bar{v},\delta,\lambda)$ is given by:*

$$\bar{c}(p,\bar{v},\delta,\lambda) = p\left[1 - (1-\delta)F(\frac{\bar{q}}{\bar{G}(2p-\delta p)}) + \delta F(\frac{\bar{q}}{\bar{G}(\delta p)}) - (1-\delta)\frac{\partial \bar{G}(2p-\delta p)}{\partial q}\int_0^{\frac{\bar{q}}{\bar{G}(2p-\delta p)}} x\mathrm{d}F(x)\right];$$

*(b) The joint solutions $(q^*, \hat{v}_0^*)$ and the profit $\Pi^*$ are increasing in $\lambda$.*

Proposition 5 shows the monotonicity of $q^*$, $\hat{v}_0^*$, and profit $\Pi^*$ with respect to the behavioral preferences of strategic customers. From this proposition, we infer that the order quantity should be higher than in the standard model if customers are loss-averse. Furthermore, the greater the loss aversion of the customers, the larger the increase in order quantity should be. However, if strategic customers are risk-averse, whether the order quantity should be higher or lower than in the standard model depends on other relevant parameters integrated into the threshold $\bar{c}(p,\bar{v},\delta,\lambda)$. This threshold $\bar{c}(p,\bar{v},\delta,\lambda)$ is computed from $(\bar{q}(p,\bar{v},\delta,\lambda), 2p - \delta p)$, as illustrated in Proposition 3 and Fig. 3. Thus, this monotonicity change threshold can be visualized as the black line in Proposition 4(a) with $c = \bar{c}(p,\bar{v},\delta,\lambda)$ in Fig. 3, intersecting the point $(\bar{q}(p,\bar{v},\delta,\lambda), 2p-\delta p)$. This indicates that the effect of customer risk aversion on order quantity and profit is somewhat intricate. When the cost exceeds the threshold $\bar{c}$, the retailer increases the order quantity to achieve higher profits in the presence of customer risk aversion; however, when the cost is below the threshold, the retailer reduces the order quantity, resulting in lower profits. Fig. 5 (with details provided in the next subsection) exemplifies Proposition 5, demonstrating the monotonicity of the joint solution and profit.

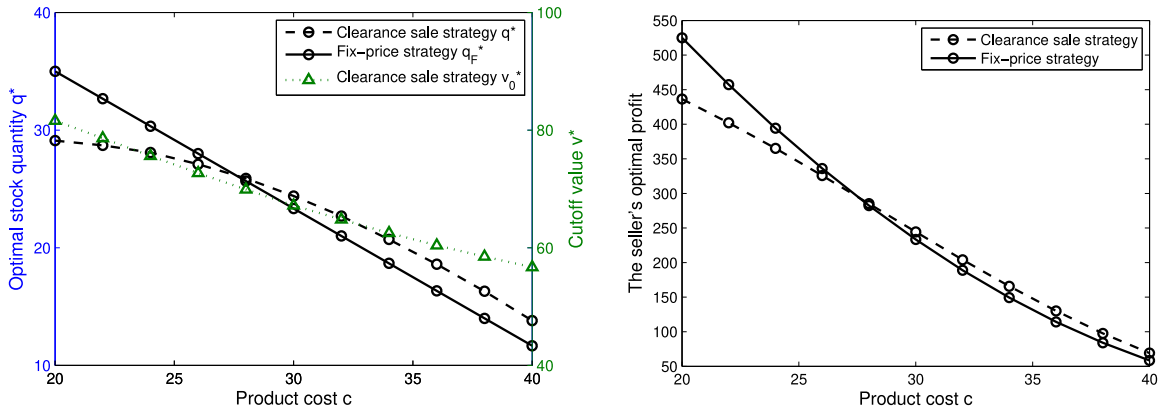



**Fig. 4.** Fixed price versus clearance sales with changing of $c$ (with $\delta = 0.5$).

### 4.3. The comparison of the two strategies

The above discussion formulates how the stock quantity can be determined under either a fixed-price strategy or a clearance sale strategy. Indeed, which strategy the seller follows critically depends on the comparison of the maximum profits. We assume a uniform distribution $x \sim unif[0, N]$, and then Proposition 6 can be presented.

**Proposition 6.** *Let* $A = \frac{N(\bar{v}-\hat{v}^*)(\bar{v}-\delta p)}{\bar{v}p(1-\delta)(\bar{v}-\delta p)-\delta\bar{v}p(\bar{v}-\hat{v}^*)}$, $B = \frac{\delta p}{2N\bar{G}(\delta p)}$, $D = \frac{N(\bar{v}-p)}{2\bar{v}p}$, $E = \frac{(1-\delta)pN\bar{G}(\hat{v}^*)}{2}$, *and let* $\Delta = (\delta pA + pA - 2pAB - 2pD)^2 - 4(A - AB - D)[p^2(\delta A - AB - D) + E]$.

*If* $\Delta > 0$, *then a clearance sale strategy is preferable to a fixed-price strategy when:*

$$c > \frac{(\delta pA + pA - 2pAB - 2pD) + \sqrt{\Delta}}{2(A - AB - D)}.$$

Proposition 6 describes a qualitative condition regarding the cost of the product to determine the optimal sale strategy. Because we have to scale the inequality in the derivation of Proposition 6, the inequality by which a fixed-price strategy is better cannot be given directly. However, we can infer that a fixed-price strategy is better than a clearance sale strategy when $c$ is smaller than a certain condition which is a little bit less than the formula in Proposition 6. That is, if the cost is high enough, the clearance sale can be the better sale strategy. Conversely, if the cost is low, the fixed-price is more likely to be the better sale strategy. Given the price $p$, low cost means a high-profit margin. This result has also been proved by Su [3] under different settings, with which we may conclude that if the seller sells products with a high-profit margin, she had better not use the clearance sale strategy. In general, the lower the product cost is, the more stock quantity of the product the seller tends to place. According to Proposition 1, the increased stock quantity releases the customers' hesitation to buy immediately and turns out to backfire against the improvement of the seller's profit. Fig. 4 shows examples of Proposition 6 when $\gamma = 1$ and $\lambda = 1$.

Fig. 5 presents the optimal solution and profit changes with the behavioral parameters. The two figures in the first row are the optimal solution and profit with $\lambda = 1$ and $\gamma = 0.7$; the two figures in the second row are the optimal solution and profit with $\lambda = 1.4$ and $\gamma = 1$. There are three observations here. First, loss aversion leads to increased decisions and profits and may make the clearance sale strategy more advantageous. Second, risk aversion leads to higher decisions and profits and may make the clearance sale strategy more advantageous when the costs is large ($c > \bar{c}(p, \bar{v}, \delta, \lambda)$). Third, the negative effects of risk aversion usually occur when the cost is low ($c < \bar{c}(p, \bar{v}, \delta, \lambda)$). In such cases the fix-price strategy is often better than the clearance sale strategy, and risk aversion cannot bring about changes in the sale strategies because it only makes clearance sale strategy worse when $c < \bar{c}(p, \bar{v}, \delta, \lambda)$. Finally, although the threshold $c$ defined in Proposition 6 cannot be provided due to its complexity, the variation of the intersections in Fig. 4 is the evident that the threshold is associated with the two behavioral parameters. These observations are consistent with the prediction of Proposition 5 and also can be deduced by Fig. 3.

We next analyze how discount rate matter because it affect the cutoff value and thereby change the sale strategy that the seller follows. When the seller follows a clearance sale strategy, an increased discount rate implies reduced incentives of waiting; hence, more immediately-buying decisions are incurred, that is, a lower fill rate in period 2. Nevertheless, the high discounted price also serves as an opposite force that clears leftover products because only customers with $v > \delta p$ buy the leftovers. Fig. 6 shows examples for the optimal problems (5) and (6) regarding the discount rate when $\gamma = 1$ and $\lambda = 1$. We find that when $\delta$ is low, the fixed-price strategy dominates the clearance sales one; whereas when $\delta$ is high, the outcome is reversed. This result can be presented intuitively that if the seller can clear all the leftovers at a high discount price, the clearance sale strategy should be a good option.

As to the cases that a fixed-price dominates the clearance sale, the presence of behavioral factors of strategic customers affects and may even reverse the outcomes. This is because the behavioral factors $\gamma$ and $\lambda$ affect the cutoff value $\hat{v}^*$, and thereby change the buying or waiting behaviors (according to Eqs. (6)). Intuitively, when loss aversion and/or risk aversion are extremely high, more customers would choose to buy immediately under the clearance sale. Hence, the clearance sale strategy serves as a priority. In the next section, we explicitly discuss the impacts of behavioral factors.

## 5. Quantifying the impacts of decision biases of customers

In this section, we address the questions that under what conditions of behavioral factors will the seller changeover her sale strategy. As has been noted, we shall discuss how do the cost and discount rate interact with the behavioral factors.

### 5.1. Fixed-price or clearance sale facing customers with decision biases?

We conduct numerical experiments to examine the impacts of behavioral factors on either a fixed-price or a clearance sale strategy is optimal as well as how their impacts vary with product cost and discount rate. A large parameter set is chosen to ensure the robustness of our conclusion. As examples, we present the parameters as follows: $X \sim \text{unif}[0, 100]$, $\bar{v} = 120$, $p = 50$, the cost parameter $c \in [20, 40]$ with an increment of 2, and the discount rate $\delta \in [0.3, 0.7]$ with an increment of 0.1. The ranges of behavioral factors are in accordance with the typical estimates of empirical studies. Specifically, the loss-aversion parameters range from 1.3 to 2.6 (e.g., 44); the risk aversion parameters range from 0.4 to 1 [9], both of which with an increment of 0.1. Hence,

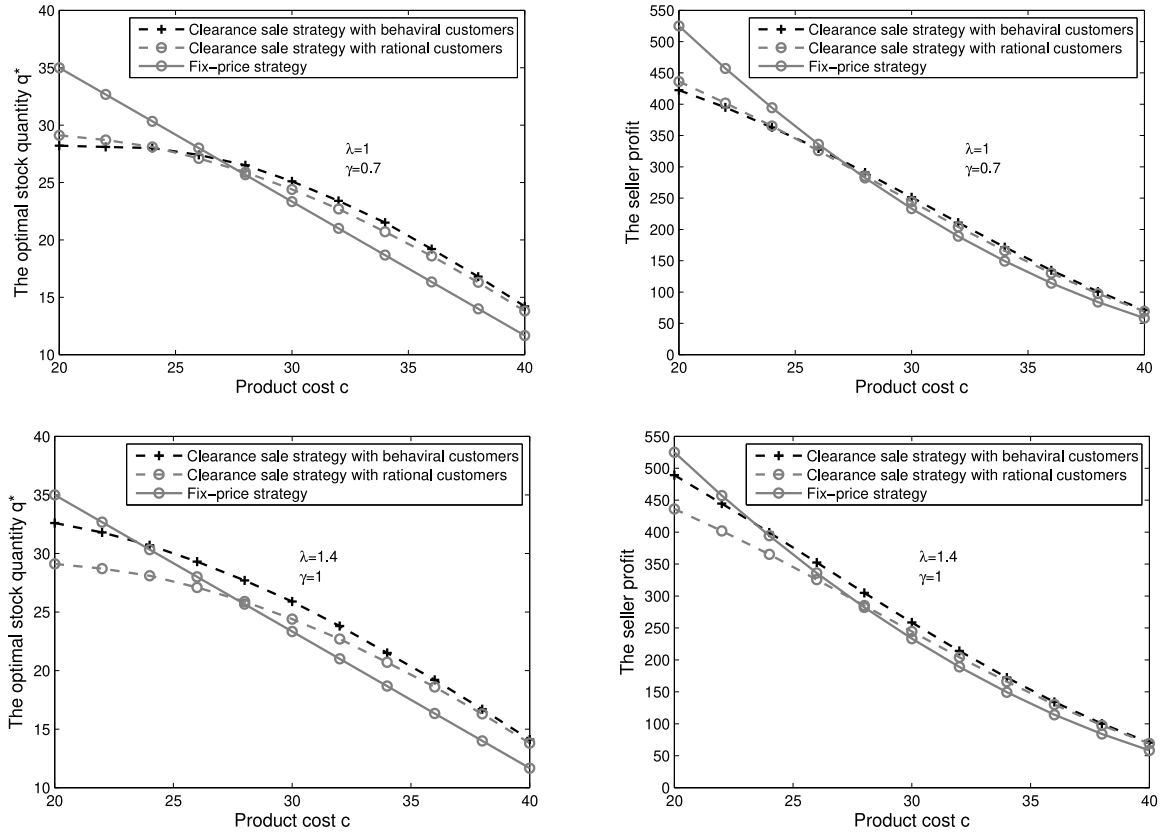



**Fig. 5.** Optimal solution and profit changes with the behavioral parameters (with $\delta = 0.5$).

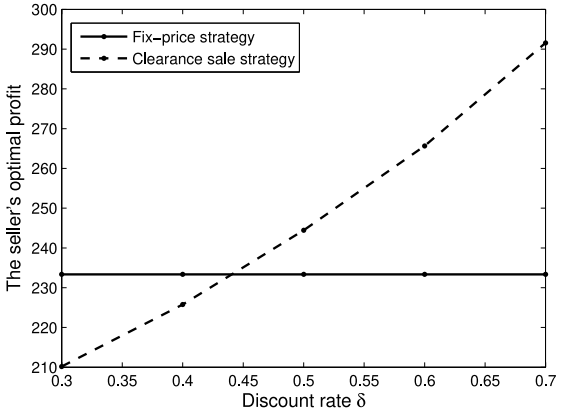



**Fig. 6.** Fixed price versus clearance sales with changing of $\delta$ (with $c = 30$).

there are a total of 5390 parameter combinations. To evaluate the optimality of the sale strategy, we compute the optimal profits under each parameter combination when the seller follows either a fixed-price or a clearance sale strategy, and then compare the corresponding profits to see which strategy generates larger profits. Fig. 7 summarizes the optimality of fixed-price versus clearance sale regarding the strength of the behavioral factors. The horizontal axis refers to risk aversion, where a large $\gamma$ indicates a low level of risk aversion. The vertical axis shows the strength of loss aversion. The region is divided by several curves of ($\gamma$, $\lambda$), at which two sale strategies are indifferent. The region below (above) the indifference curve indicates the optimality of a fixed-price (clearance) sale strategy.

There are several observations here. First, both risk aversion and loss aversion have a significant impact on the seller's decision of the sale strategy. When customers are more influenced by the risk or the disutility of obtaining nothing due to stock out (moving left horizontally), the clearance sale outperforms the fixed price to more often be the seller's optimal strategy. That is, compared to models that do not capture risk aversion and loss aversion, the presence of those behavioral factors encourages the use of clearance sale. Second, the product cost divides the region delicately. As seen, the indifference curve moves downward as the product cost increases. This means that a low product cost mitigates the leverage effect of behavioral factors and discourages the use of the clearance sale strategy. Thus, even with (limited levels of) behavioral factors, industries with low product costs (e.g., branded pastries) will still persist in following the fixed price. Third, the discount rate also has consequences on the optimality of the sale strategy. When the discount rates are anticipated to be very low, the seller prefers the fixed-price strategy. This is because: The increased discount rate reduces a customer's utility of waiting and thus suppresses the strategic behaviors of customers. The seller is able to retain certain proportions of customers buying at the full price in period 1 but also salvage the leftovers in period 2 so that her profit is improved. In this sense, the seller should be more concerned about setting a proper discount rate when behavioral factors are at play if she has pricing power. Özer and Zheng [10] suggested the discount rate should fall in the range of $[0.4, 0.8]$. We also observe asymmetric regions ($\gamma$, $\lambda$) that the clearance sale strategy outperforms the fixed-price one over broader ranges of behavioral factors.

### 5.2. The impacts of specific behavioral factors

The previous subsection presents the impact of behavioral factors on whether to adopt either a clearance or fixed price sale strategy. Two questions arise as to how the behavioral factors individually impact the seller's profit and how they interact with each other under a clearance sale. To answer these questions, we follow a similar approach

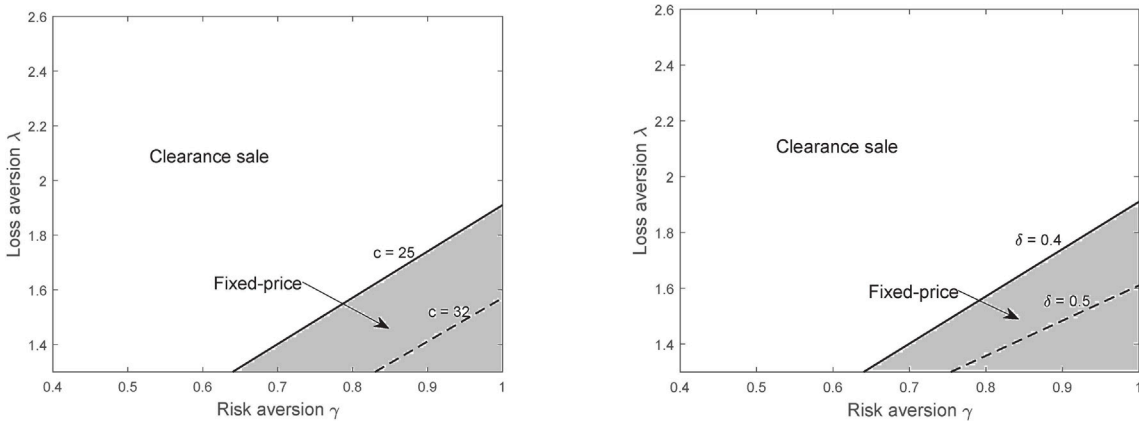



**Fig. 7.** Clearance sale vs. fixed-price regarding the cost (left) and the discount rate (right).

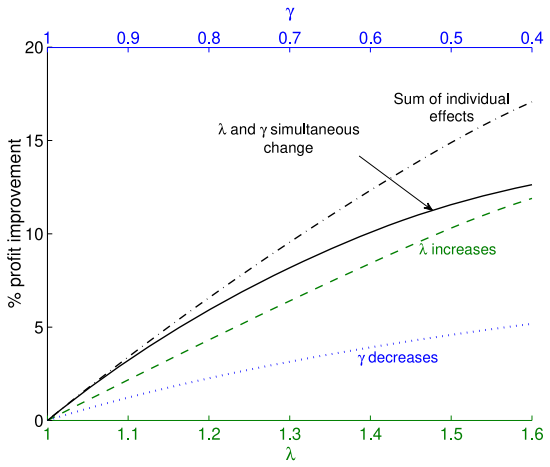



**Fig. 8.** Profit improvement (%) with respect to $(\lambda, \gamma)$ with $\delta = 0.5$ and $c = 30$.

**Table 1**
Statistics for differences of profit (%) and stock quantity (%) due to incorporating behavioral factors.

| | Discount rate $\delta$ | | | | | |
|---|---|---|---|---|---|---|
| | 0.2 | | 0.4 | | 0.6 | |
| | $\Pi$ | $q$ | $\Pi$ | $q$ | $\Pi$ | $q$ |
| Product cost $c = 20$ | | | | | | |
| Average | 11.12 | 38.67 | 9.36 | 34.97 | 12.70 | 45.47 |
| Std.Dev | 4.32 | 12.24 | 3.10 | 9.62 | 1.82 | 6.33 |
| Max | 18.55 | 60.36 | 14.57 | 51.61 | 15.49 | 55.65 |
| Product cost $c = 30$ | | | | | | |
| Average | 4.69 | 24.58 | 2.53 | 17.24 | 0.63 | 8.21 |
| Std.Dev | 1.86 | 7.06 | 0.97 | 4.60 | 0.23 | 1.88 |
| Max | 7.67 | 35.48 | 4.09 | 24.53 | 0.99 | 10.92 |
| Product cost $c = 40$ | | | | | | |
| Average | 0.84 | 9.58 | 0.33 | 5.92 | 0.09 | 2.92 |
| Std.Dev | 0.37 | 2.63 | 0.15 | 1.67 | 0.03 | 0.53 |
| Max | 1.38 | 14.04 | 0.56 | 8.06 | 0.13 | 3.13 |

$\Pi$ and $q$ denote the dimension of the profit and the stock quantity.
% difference = (optimal value - suboptimal value)/suboptimal value × 100%.

with Özer and Zheng [10], which is summarized as follows. First, the "optimal profit" under perfect rationality where $\gamma = 1$ and $\lambda = 1$ are seemed as a benchmark profit and compute the optimal profit and corresponding improvement of the profit when $\gamma$ decreases (i.e., more risk averse) or $\lambda$ increases (i.e., more loss averse). Second, we sum up the profit improvements. Here, the profit improvement is defined as the difference between the optimal profit and the benchmark profit divided by the benchmark profit. Finally, the profit and the corresponding improvement by jointly changing $\lambda$ and $\gamma$ in equal magnitudes are computed, and then we compare the "joint" improvement of profit to the sum of profit improvements. In doing so, we are able to examine the regularity with respect to each behavioral parameter; more importantly, to see whether the two behavioral parameters reinforce or mitigate each other on the seller's profit.

Fig. 8 shows the representative results. The solid line shows the sum of the percentage profit changes due to individual improvement of each behavioral parameter. The dashed line shows the percentage improvement in the seller's optimal profit when both behavioral parameters simultaneously change. From the figure, we find that the profit is improved with the increase of $\lambda$ and the decrease of $\gamma$. Moreover, the profit improvement associated with loss aversion is higher than that associated with risk aversion in this case, but this result may change with different parameter setting. We also see that the profit improvement due to the joint change of $\lambda$ (increases) and $\gamma$ (decreases) (the dash–dot line) is substantially lower than the sum of profit increases when either parameter changes (the solid line). That is, risk aversion and loss aversion mitigate each other in improving the seller's profit. The potential reason is that risk averse and loss averse customers are more prone to buy the product immediately at the full price (for any given quantity of products).

### 5.3. Decision and profit impacts when incorporating behavioral factors

Recognizing the profit implication of the behavioral factors, one of the most important concerns for a seller is how much profit (improvement) will be attained if the behavioral factors are fully incorporated into the decision. To answer this question, the numerical experiments are conducted by the following three steps. First, we compute a seller's rational "optimal" stock quantity and the "optimal" profit when the seller ignores the behavioral factors, that is, assuming that $\gamma = 1$ and $\lambda = 1$. Second, given the "optimal" stock quantity, we compute the seller's resultant profit when customer purchase decisions are in fact affected by the ignored behavioral factors. We denote the "optimal" stock quantity and the computed profit in this condition as the suboptimal stock quantity and the suboptimal profit. Finally, we compute the optimal stock quantity and the optimal profit when the seller is aware of and fully incorporates the behavioral factors. The above computations are repeated for each combination of factors involving $\gamma$, $\lambda$, $c$, and $\delta$. Thus, the percentage profit (and stock quantity) improvement can be calculated, which is defined as the difference between the optimal profit (and stock quantity) and the suboptimal one divided by the suboptimal one.

Same as the previous parameters setting: $X \sim \text{unif}[0, 100]$, $\bar{v} = 120$, $p = 50$, $\gamma \in [0.4, 1]$, $\lambda \in [1.3, 2.6]$, both $\gamma$ and $\lambda$ with an increment

of 0.1. For ease of exposition, we select three levels of product costs (i.e., $c = 20$, 30, and 40) and three levels of discounts (i.e., $\delta = 0.3$, 0.5, and 0.7). Hence, there are a total of 882 parameter combinations. Table 1 shows the statistics for the improvements of the profit and the stock quantity for each combination $(c, \delta)$. Incorporating the behavioral factors can generate as high as 18.55% profit improvement. The potential profit improvement is due to the increased stock quantity. Actually, this improvement is remarkable because the average profit margins of industries frequently using clearance sales for grocery and nonbranded apparel vary between 1% and 8% [10]. A 18.55% profit improvement is a significant difference. We note that the profit improvements for low-cost products in our experiment are higher than that for average improvement as predicted by Özer and Zheng [10] (3.87% versus 2.42%).

We also find that the profit improvement is closely related to the product cost and the discount rate. For any $\delta$, as the product cost increases, the profit improvement sharply decreases. Nevertheless, how the discount rate $\delta$ impacts the improvement depends on the product cost. In the case of a low cost ($c = 20$), the profit improvement first decreases then increases (from 11.12% to 9.36% to 12.70%) as $\delta$ increases from 0.3 to 0.5 to 0.7. In the case of a moderate or high cost ($c = 30$ or 40), the profit improvement decreases as $c$ increases. In such an extreme case as $c = 40$ and $\delta = 0.7$, incorporating behavioral factors generates little profit improvement (0.09%). There are two reasons that account for such limited consequence. First, when the product cost is high, the seller is likely to prepare less quantity of products, which implies a lower fill rate. Second, a high discount means a smaller utility of waiting. These two reasons together drive customers to buy the product immediately. Thus, thinking more about the product cost and the discount rate adds insight for a clearance sale strategy. Our conclusion emerges as a compelling reason for a seller to provoke consumer behavioral factors when the product cost and the discount rate are not too high, thereby improving her profitability.

### 5.4. Decision and profit impacts when misperceiving behavioral factors

Individuals exhibit heterogeneous behavioral factors [45,46] and the strengths of behavioral factors may also vary with different conditions. For example, experimental results have shown that the scarcity of products [9] and a small payment scale [45,47] decreases the strength of risk aversion; a high price is likely to induce a low level of loss aversion [42]. Consequently, it is commonplace for the seller to misperceive the strength of behavioral factors. To assess the value of accurately estimating the behavioral factors, we examine the profit and quantity decision impacts due to such misperception. Specifically, we assume the true values of $\delta$ and $\lambda$ to be the midpoints of the experimental estimates, that is, $\delta = 0.7$ and $\lambda = 1.95$. The seller may either overestimate or underestimate each of the parameters. Then, the stock quantity of products when misperceiving the behavioral factors and the resultant profit (as the misperceived profit) can be calculated. We similarly define the profit improvement as the difference between the optimal profit under accurate estimates of behavioral factors and the misperceived profit divided by the misperceived profit.

We specifically look at how the impacts of misperception vary with the product cost and the discount rate. Again, we choose $c = 20$, 30, and 40, $\delta = 0.2$, 0.4, and 0.6. Table 2 demonstrates the results. It can be observed that the misperception of behavioral factors can cause either less or more stock quantity, thereby reducing the seller's profit. The average profit improvement is at most 2.02%, which occurs when the product cost and the discount rate are low. As the product cost and the discount rate increase, the average improvement decreases and even vanishes. In that case, compared to the 5.96% profit improvement due to complete ignorance of the behavioral factors, the profit consequence of misperceiving the behavioral factors is much smaller. Still, the maximum profit improvement can be as high as 57.29% when the product cost is low and the discount rate is moderate. This conclusion is in sharp contrast to the study of Özer and Zheng [10], which suggests that the maximum profit is capped at 1.5%. Actually, from our experiment, we observe that only when the product cost is high, or the product cost is moderate but the discount rate is high, the maximum profit improvement is capped at 1.5%. This, in turn, gives supports that product cost and discount rate play an important role in evaluating the impact of behavioral factors.

**Table 2**
Statistics for differences of profit and quantity (%) due to misperceiving behavioral factors.

| | Discount rate $\delta$ | | | | | |
|---|---|---|---|---|---|---|
| | 0.2 | | 0.4 | | 0.6 | |
| | $\Pi$ | $q$ | $\Pi$ | $q$ | $\Pi$ | $q$ |
| Product cost $c = 20$ | | | | | | |
| Average | 2.02 | −0.05 | 1.71 | −0.38 | 0.60 | −0.34 |
| Std.Dev | 4.90 | 8.82 | 6.14 | 7.10 | 2.44 | 4.34 |
| Max | 39.62 | −20.78 | 57.29 | −17.86 | 23.30 | −12.15 |
| Product cost $c = 30$ | | | | | | |
| Average | 0.44 | −0.12 | 0.21 | −0.58 | 0.03 | −0.18 |
| Std.Dev | 0.73 | 8.82 | 0.35 | 3.90 | 0.05 | 1.73 |
| Max | 4.75 | −20.78 | 2.24 | −11.20 | 0.35 | −5.43 |
| Product cost $c = 40$ | | | | | | |
| Average | 0.07 | −0.86 | 0.03 | −0.49 | 0.00 | −0.20 |
| Std.Dev | 0.12 | 2.38 | 0.04 | 1.57 | 0.00 | 0.52 |
| Max | 0.71 | −7.94 | 0.25 | −4.55 | 0.03 | −1.52 |

$\Pi$ and $q$ denote the dimension of the profit and the stock quantity. Given $(c, \delta)$, the average of optimal profits under a clearance sale is larger than that under a fixed price. Difference = (optimal value - misperceived value)/misperceived value × 100%. The negative (positive) number indicates a decrease (increase) in profit or quantity due to misperception.

The above results provide insight for a seller to manage clearance sales. In practice, obtaining accurate estimates of behavioral factors may cost thousands of dollars. A seller has to consider upon the potential profit improvement against the relevant cost. Our analysis implies that industries of high product costs or moderate product costs with high discounts can take less care about the inaccurate estimates. On the contrary, industries of low product costs that offer low discounts are suggested to never tire of mining customer preferences. This is accessible nowadays because of recent advances in information technologies (e.g., data mining).

### 5.5. Decision and profit impacts when misperceiving behavioral factors

Individuals exhibit heterogeneous behavioral factors [45,46] and the strengths of behavioral factors may also vary with different conditions. For example, experimental results have shown that the scarcity of products [9] and a small payment scale [45,47] decreases the strength of risk aversion; a high price is likely to induce a low level of loss aversion [42]. Consequently, it is commonplace for the seller to misperceive the strength of behavioral factors. To assess the value of accurately estimating the behavioral factors, we examine the profit and quantity decision impacts due to such misperception. We assume the misperception value of behavioral factors are a range in accordance with the typical estimates of empirical studies: the loss-aversion parameters range from 1.3 to 2.6 (e.g., 44); the risk aversion parameters range from 0.4 to 1 [9], both of which with an increment of 0.1. Specifically, we assume the true values of $\gamma$ and $\lambda$ to be the midpoints of the experimental estimates, that is, $\gamma = 0.7$ and $\lambda = 1.95$. Accordingly, the seller may either overestimate or underestimate each of the true value of $\gamma$ and $\lambda$. Then, the stock quantity of products when misperceiving the behavioral factors and the resultant profit (as the misperceived profit) can be calculated. We similarly define the profit improvement as the difference between the optimal profit under accurate estimates of behavioral factors and the misperceived profit divided by the misperceived profit.

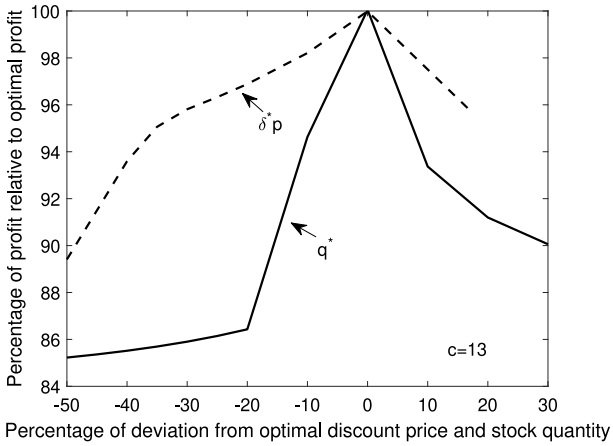



**Fig. 9.** Sensitivity of profits to errors in the price and stock quantity.

We specifically look at how the impacts of misperception vary with the product cost and the discount rate. Again, we choose $c = 20$, 30, and 40, $\delta = 0.3$, 0.5, and 0.7. Table 2 demonstrates the results. It can be observed that the misperception of behavioral factors can cause either less or more stock quantity, thereby reducing the seller's profit. The average profit improvement is at most 2.02%, which occurs when the product cost and the discount rate are low. As the product cost and the discount rate increase, the average improvement decreases and even vanishes. In that case, compared to the profit improvement due to complete ignorance of the behavioral factors, the profit consequence of misperceiving the behavioral factors is much smaller.

The above results provide insight for a seller to manage clearance sales. In practice, obtaining accurate estimates of behavioral factors may cost thousands of dollars. A seller has to consider upon the potential profit improvement against the relevant cost. Our analysis implies that industries of high product costs or moderate product costs with high discounts can take less care about the inaccurate estimates. On the contrary, industries of low product costs that offer low discounts are suggested to never tire of mining customer preferences. This is accessible nowadays because of recent advances in information technologies (e.g., data mining).

Note that the numerical analysis is based on the assumption of uniform distribution of both customers' demand and value of products. In order to verify the stability of the conclusions, we also run the numerical analysis under the normal distribution assumption. The corresponding content can be seen in Appendix B.

## 6. Sensitivity of seller's profits to price and stock quantity decisions

In the previous study, price is given exogenously, we did not consider the impact of the price on the seller's profit. In this section, we investigate whether profits are more sensitive to the seller's price or stock quantity decision when facing decision biases of customers. To analyze this issue, we take discount price as one of the seller's decisions:

$$\begin{aligned}\max\ \ & \Pi(q,\delta,\hat{v}^*) = pE\Big[\bar{G}(\hat{v}^*)X \wedge q\Big] \\ & + \delta pE\Big[(q-\bar{G}(\hat{v}^*)X)^+ \wedge \big(G(\hat{v}^*)-G(\delta p)\big)X\Big] - cq, \\ s.t.\ \ & \frac{\lambda(\hat{v}^*-p)^\gamma}{(p-\delta p)^\gamma+\lambda(\hat{v}^*-p)^\gamma} = F\Big(\frac{q}{\bar{G}(\delta p)}\Big) \\ & + \int_{\frac{q}{\bar{G}(\delta p)}}^{\frac{q}{\bar{G}(\hat{v}^*)}} \frac{q-x\bar{G}(\hat{v}^*)}{x[\bar{G}(\delta p)-\bar{G}(\hat{v}^*)]} f(x)dx. \end{aligned} \tag{7}$$

Given price $p$ in period 1, we first numerically found the optimal discount price $\delta^* p$, stock quantity $q^*$ and the corresponding profit. Next, we respectively computed the relative profit losses under the conditions that the discount price and stock quantity deviate from the optimal decisions, which can be expressed as $(1-\frac{\Pi}{\Pi^*})100\%$. Finally, we compare the relative profit losses, i.e., the sensitivity of the seller's profits to discount price and stock quantity decisions.

We present a numerical example with $\bar{v} = 120$, $p = 50$, $c = 30$, $N = 100$, $\gamma = 0.7$, $\lambda = 1.4$. In this example, as Fig. 9 shows, when the discount price deviates from the optimal discount price by 30%, a relative profit loss of 5% is caused. When the stock quantity decision deviates from the optimal inventory decision by 30%, a relative profit loss is as high as 14%. We can observe that profit is more sensitive to the error of stock quantity decisions. This result suggests that the seller's stock quantity decision may be more important than the price decision when facing strategic consumers with decision biases. Hence, it is necessary to focus on the stock quantity decision of the seller principally.

## 7. Conclusions

It is important for a seller to optimize its sale and stocking quantity decisions when facing a stochastic number of strategic customers. In this study, we consider the situation in which customers are risk averse and loss averse, and analyze how the seller's decisions are leveraged by the two specific factors. We show that when customers are risk and loss neutral, a clearance sale could be more profitable than a fixed-price strategy under some conditions associated with product cost and a discount. This conclusion is in consistent to what is reported in the literature. We also find that risk aversion does not always favor the adoption of a clearance sale strategy, although loss aversion does. To be specific, when the discount and product costs are relatively high, risk aversion tends to favor a clearance sale strategy. While the discount and product costs are low, risk aversion may do the opposite.

In addition, we highlight the potential benefits of considering multiple behavioral factors. Numerical examples suggest that under a clearance sale strategy, a seller can gain as high as 18.55% potential profit improvement due to considering the behavioral factors. Such profit improvements are closely associated with the product cost and discount. Specifically, high-cost products have much less profit improvements compared to low-cost ones. Hence, in the former cases, costly efforts of leveraging and estimating behavioral factors may not be necessary. We also point out that inaccurately estimating the behavioral factors can result in great profit losses, but when the product cost is high or the product cost is moderate with a high discount rate, a seller need not worry excessively about the inaccuracy because it causes limited profit consequences.

In summary, our result provides behavioral explanations for contrasting sale strategies among luxury companies such as LV and Gucci, fashion-branded companies such as Zara and H&M and warehouse wholesalers such as Costco and Sams. In general, luxury goods with low cost (high-profit margin) is almost no discount. However, branded fashion products are likely to induce strong loss aversion and risk aversion, favoring a clearance sale strategy. Moderate-profit margin enhances such tendency. In contrast, every-use products that Costco and Sams mainly offer tend to incur weak loss aversion and risk aversion. Low-profit margin enhances such tendency. As a result, a fixed-price strategy is applied to those products. Hence, we suggest that when making the decisions of sale strategy and order quantity, a seller should comprehensively consider the behavioral factors, and also product cost and discount rate.

There are several directions for future research. First, as has been noted, an increased number of products incurs a high strength of risk aversion [9]. When taking this relation into account, a seller can benefit more from the behavioral factors. After that, the condition for each sale strategy and the consequential optimal stock quantity may differ and need to be further addressed. Second, for ease of analysis, we assume a uniform distribution of customer valuations and demand. Because both

of these are associated with stockout, surveying actual distributions add value to assess the benefits due to behavioral factors [26]. Finally, exploring more intricate competitive market models is essential for enhancing the external validity and practical relevance of the study.

## CRediT authorship contribution statement

**Yanan Song:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Dong Xie:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **Wei Gu:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition, Formal analysis, Conceptualization.

## Declaration of competing interest

We confirm that this work is original and has not been published elsewhere, nor is it currently under consideration for publication elsewhere. We have no conflicts of interest to disclose.

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## Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.omega.2024.103265.

## Data availability

No data was used for the research described in the article.

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